



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University

NAAC Accredited with 'A+' Grade & An ISO 9001: 2015 Certified Institution

NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Engineering

Academic Year 2025– 2026 (Odd Semester)

Degree, Semester & Branch : V Semester B.E CSE
Course Code & Title : CCS366 & Software Testing and Automation
Name of the Faculty member : Mrs. S. Vijaya Amala Devi, AP / CSE

Innovative Practice Description

- **Unit / Topic:** Unit IV/Advanced Testing Concepts/Security Testing
- **Course Outcome:**4
- **Topic Learning Outcome:** 4b
- **Activity Chosen:** Reflection
- **Justification:**
 - Security Testing is a complex topic involving concepts such as authentication, authorization, input validation, session management, SQL injection, XSS, CSRF, and vulnerability scanning. Through reflection, it was evident that peer discussions and collaborative demonstrations helped students internalize these concepts more effectively than traditional lecture-based teaching.
 - They were able to identify security weaknesses, interpret system behavior during attacks, and understand the critical role of defensive coding practices. The practice also enhanced their problem-solving abilities, communication skills, and professional readiness in cybersecurity. Thus, the reflective outcomes strongly justify the effectiveness and necessity of implementing innovative teaching methods for this topic.
- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
 - The implementation of the innovative practice for the topic **Security Testing** began with an introduction to essential cybersecurity concepts, including vulnerabilities, threats, authentication, authorization, input validation, and common cyberattacks such as SQL Injection, XSS, and CSRF. Students were provided with pre-class materials, which allowed them to review these topics in advance and build a strong conceptual foundation.
 - During the session, the peer teacher guided the class through simplified explanations of key security principles and demonstrated sample attack scenarios using basic tools. This step-by-step approach enabled students to clearly understand how vulnerabilities occur and how preventive measures ensure the protection of software systems. The hands-on demonstrations made the learning experience practical and helped students visualize real-world security issues.
 - Following the demonstrations, students engaged in group discussions where they analyzed potential security threats, predicted system weaknesses, and proposed suitable mitigation strategies. This collaborative activity strengthened their critical thinking, communication, and teamwork skills.

- Students Ms. Shenbagapriya and Ms. Sathyapriya reflected that applying security concepts to real-time applications, such as e-commerce platforms, helped them connect theoretical knowledge with practical implementation, thereby improving their appreciation of secure software development practices.
- Figure 1 shows students reflecting on their understanding of the concepts. Faculty feedback at the end of the session clarified misunderstandings and reinforced major learning points. Overall, the peer learning-based innovative practice enhanced learner confidence, improved conceptual clarity, and deepened their awareness of the significance of security testing in the software development lifecycle.

CO – PO / PSO mapping:

CO	PO1	PO2	PO5	PO8	PO10	PO12	PSO3
CO4	3	3	3	3	3	3	3

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO4	PO5	PO10	PO12	PSO2
	3	3	3	1	2	2	2
Justification for correlation	Student demonstrating testing practices involves applying basic engineering principles to understand and testing concepts and methodologies	Student carrying out advanced types of testing requires analyzing complex problems and identifying suitable testing techniques to address them effectively.	Student advanced testing practices involve investigating complex issues and applying advanced testing techniques to uncover underlying defects and vulnerabilities.	Student demonstrating testing practices includes utilizing modern testing tools and technologies to automate testing processes and enhance testing efficiency.	Student effective communication of testing practices and results to stakeholders ensures alignment and support for testing efforts, contributing to successful project outcomes.	Student demonstrating testing practices requires a commitment to continuous learning to stay updated with evolving testing methodologies and technologies.	Students can demonstrate the ability to ensure the reliability and security of IT solutions, aligning with the application of IoT, cloud computing, and cybersecurity knowledge.

- **Images / Screenshot of the practice:**



Fig 1: Student discussed the topics

Reflective Critique:

- **Feedback from Students and Stakeholders:**

- Students shared that the peer learning activity on **Security Testing** helped them understand complex security concepts more clearly and confidently. They appreciated the pre-class materials, which allowed them to review important topics such as vulnerabilities, authentication, authorization, SQL Injection, XSS, CSRF, and other common cyberattacks at their own pace.
- During the peer-led session, students found it easier to clarify doubts and strengthen their understanding through discussions and live demonstrations. Stakeholders also noted increased engagement, as students were able to connect theoretical concepts with real-world cybersecurity challenges in modern applications.

- ❖ **Benefits of the Practice:**

- Peer learning improved communication, collaboration, and teamwork among students.
- Explaining concepts such as vulnerability scanning, penetration testing, and secure coding strengthened students' own understanding.
- Hands-on demonstrations helped students connect theory with real-world security scenarios.

- Students developed stronger analytical and problem-solving skills by analyzing attack patterns and proposing mitigation strategies.
- The activity increased motivation and awareness of the importance of security in modern software applications.

❖ **Challenges Faced:**

- Some students were hesitant to become peer teachers due to the technical depth of the topic.
- A few students struggled with understanding advanced security terms such as session hijacking, CSRF, and non-repudiation.
- Students needed additional guidance to grasp complex attack mechanisms and security tools.
- Limited prior exposure to cybersecurity concepts made some learners dependent on faculty support.
- Ensuring equal participation among all students was challenging due to varying confidence levels.

References:

- ❖ https://en.wikipedia.org/wiki/Peer_learning
- ❖ [Peer to Peer Learning - Examples, Benefits & Strategies](#)
- ❖ <https://whatfix.com/blog/peer-to-peer-learning/>

Signature of Faculty Member

HOD